Programming Assignment 2 CNN and data competition

In this assignment we will construct deep neural networks for image processing using CNN.

Part 1

Use the cifar10 dataset included with keras and construct a CNN to predict the image labels. Don't forget to visualize some images to understand the data. Train the network and visualize accuracy over epochs and check the output with outputing some predictions for visual comparison.

Part 2

Use the cifar100 dataset included with keras and repeat the experiment in Part 2.

Part 3

Use the Animal10 dataset in moodle to repeat the experiment. See also the Kaggle page.

a) Download images to your computer and load the data organized as training and testing data (use 20% for testing). Explore and visualize some of the data to gain a better understanding of the problem. You can use tensorflow's image_dataset_from_directory function and other standard functions to scale the dataset to the same input sizes, etc...

b) Explore the dataset. Visualize some images, check image sizes, number of images in each folder.

c) Perform operations to achieve a balanced dataset (use undersampling or data augmentation).

d) Construct a Convolutional Neural Network to predict the correct label, and train it. Tune the structure and hyperparameters for best accuracy. Visualize accuracy over the training epochs.

Grading:
Part 1: 30%
Part 2: 30%
Part 3: 30%
Accuracy score on Part 3: 10%

- Baseline: 60% validation accuracy. You need to achieve at least 60% accuracy with your CNN.
- Highest accuracy in the class on 20% validation data = 10%
- Lower accuracy will be awarded points proportionally. e.g. 90th percentile will get 9%, etc.